

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Database Management System**

Code No: **CACS255** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is DBMS? Describe the advantages and disadvantages of using DBMS.
3. Describe 3-schema architecture of database.
4. What are roles and responsibilities of DBA.
5. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.
6. What are the states of transaction model? Describe the ACID properties of transaction.
7. Describe the use of ER diagram. Draw an ER diagram for a university to design a database for it.
8. What is DDL and DML queries? Consider the following relation.

Group C

Attempt any TWO questions[2x10=20]

9. Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples.
10. Differentiate between stored procedure and trigger. Why concurrency control is needed? Describe timestamp based concurrency protocol.
11. What is data model? What are its types? Describe the structure of database.